Ranking Grid Cells for Landmine Survey in Antioquia, Colombia

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Overview

In this project I built and tested supervised machine learning models that estimate landmine risk for 500 × 500 m grid cells, so that the cells of a municipality can be ranked for survey. I used the RELand landmine-risk dataset, *Landmine Detection in Antioquia, Colombia* (mdulcer10, 2026), available at https://huggingface.co/datasets/mdulcer10/landmine-detection, which covers 26,761 cells in 15 municipalities of Antioquia that have since been declared mine-free and marks the 430 cells where demining found mines. I trained two binary classifiers, a logistic regression and gradient-boosted trees, and tested them as such a model would be used, on municipalities they had never seen, against a simple rule that surveys first the cells nearest a recorded past landmine event. The question matters to me because, from 25 years in humanitarian mine action and arms control, survey is the quick, reconnaissance-like stage that decides where the slow and resource-heavy work of clearance goes, and my goal is a digital survey: an information tool that helps to plan even the surveys more efficiently.

Dataset Description

The dataset was published on Hugging Face under a CC BY 4.0 licence with the RELand study of landmine risk in Colombia (Dulce Rubio et al., 2024), and I used its single data file, unchanged, downloaded on October 18, 2026. It has 26,761 rows, one per grid cell, and 67 columns: two coordinate columns, the municipality, the target and 63 features. The target, `mines_outcome`, is 1 for the 430 cells (1.6%) that intersect a confirmed hazardous area or a cleared area where landmines were found during humanitarian demining, using data from Colombia's national mine action authority (AICMA), and 0 for cells that were inspected and declared free of mines (mdulcer10, 2026). The 63 features are open data: 27 continuous

measures, among them distances to roads, settlements, coca crops and the nearest recorded past landmine event (`dist_old_mine`), terrain, climate, population and night-time light, eight of which have a single value for a whole municipality, and 36 one-hot columns for land use, climate zone and relief. The file has no missing values or repeated rows, and I used the coordinates only to count how clustered the mine cells are before dropping them. The dataset is hosted on Hugging Face rather than one of the sources the brief lists, but it comes with its licence, its peer-reviewed study and a recommended testing protocol, and its labels come from Colombia's national mine action records.

Modeling Approach

The notebook, data and environment file are in a public GitHub repository, https://github.com/travisrcarlson/applied-machine-learning-project.

Preparing the Data

The task is supervised binary classification: predict `mines_outcome` from the features. The municipality is not a feature; it defines the groups of cells that are held out together in testing. There were no missing values to fill and no text categories to encode, since the dataset already one-hot encodes its three categorical variables. For the logistic regression, which adds up weighted feature values, I applied log(1 + x) to the 13 distance features and to cell population and night-time light. Eight of these 15 features have long right tails, with skewness of about 1 to 8.3, which would let a few extreme cells dominate the fit; the log reduces their skewness to between −0.2 and 1.2, but it gives the seven that were barely skewed a left skew, a cost I accepted to keep every distance in the same form. I then standardised all 27 continuous features to a mean of 0 and a standard deviation of 1, so that the penalty treats them equally and their weights can be compared. Both steps sit inside a scikit-learn pipeline (Pedregosa et al., 2011), so

in every round they are learned from the training municipalities only. The trees took the raw values, since a tree splits each feature at thresholds and a transformation that keeps the order of the values leaves its splits unchanged. Because only 1.6% of cells hold mines, both models weighted the two classes so that the mine cells count as much in total as the clear cells; over the whole dataset, each mine cell weighs about 31 and each clear cell about 0.51. Finally, I trained every model twice, with all 63 features and with the 62 left without `dist_old_mine`, because that feature separates mine cells from clear cells far better than any other and may share a source with the labels (see Limitations).

Choice of Models

I chose two models that differ in what they can learn. Logistic regression is a standard and interpretable baseline: it estimates the log-odds of a mine cell as a weighted sum of the features, with an L2 penalty whose strength, `C`, I tuned. Gradient-boosted trees add up many small decision trees, each fitted to the errors of those before it, so they can capture thresholds and interactions that a linear model misses. I used scikit-learn's histogram-based implementation, which its documentation describes as inspired by LightGBM (scikit-learn developers, n.d.; Ke et al., 2017), tuned its learning rate, and kept its defaults of at most 100 trees of up to 31 leaves, with early stopping. Both models also match baselines in the RELand study (Dulce Rubio et al., 2024), whose results give a reference point. I compared them with two rankings that learn nothing: a random order, which shows what chance achieves, and a distance rule that ranks cells by nearness to the nearest recorded past landmine event, which shows what the existing records achieve without a model. A trained model is only worth using if it beats both.

Testing on Held-Out Municipalities

Mine cells cluster: 372 of the 430 (86.5%) have another mine cell among their eight neighbours. With spatially structured data, a random split puts neighbouring cells from the same contaminated area on both sides and overstates how well a model will do in a new place, so the test should hold out whole blocks (Roberts et al., 2017). I therefore held out one municipality at a time, the protocol the dataset card suggests and the RELand study uses (Dulce Rubio et al., 2024). In each of 15 rounds, I tuned the model by 3-fold cross-validation within the other 14 municipalities, keeping each municipality whole within a fold, trained it on all 14, and scored the 15th, which played no part in any choice. As a check on how much an easier test would flatter the models, I also ran one random 80/20 split of the cells.

Evaluation Metrics

Accuracy is of no use here: calling every cell clear is 98.4% accurate and finds no mines. A ranking is useful if it puts mine cells near the top, so my main metric was the area under the precision–recall curve (PR-AUC), computed as average precision; on data this imbalanced, precision–recall analysis shows differences that the more common ROC curve hides (Saito & Rehmsmeier, 2015), and a random ranking scores about the mine rate. I also report the ROC-AUC, the chance that a randomly chosen mine cell is ranked above a randomly chosen clear cell, for comparison with RELand, and I give both as the mean and standard deviation over the 15 held-out municipalities. As an operational measure, I flagged the top 10% of each municipality's cells and pooled the recall (the share of all mine cells flagged), the precision (the share of flagged cells that held mines) and F1, their harmonic mean.

Assumptions

The logistic regression assumes that the log-odds of a mine cell change linearly with each transformed feature and that the features' effects add up without interacting. Both models assume that what holds across 14 municipalities of Antioquia also holds in the 15th, and that the labels are right: that mine cells were mapped correctly and that clear cells, which the dataset describes as thoroughly inspected, hold no mines. The comparison with the distance rule also assumes that the recorded past events would be known before survey, which the data do not show.

Results

Table 1 sets out the scores of the six rankings on the 15 held-out municipalities. The other five rankings all beat the random order, but with all 63 features neither model beat the distance rule reliably. Their mean PR-AUC was higher than the rule's (.213 for the logistic regression and .218 for the trees, against .194), but the difference came from La Unión, whose single mine cell both models ranked first (PR-AUC 1.000) and the distance rule third (.333). Over the 12 municipalities with at least five mine cells, the rule led (.215, against .183 and .189), and it also led on ROC-AUC (.864, against .854 and .782) and on the share of all mine cells in the top 10% of cells (61.6%, against 59.1% and 53.5%), where about one flagged cell in ten held mines.

The distance rule is strong because mine cells lie close to past events (Figure 1). Taken alone, `dist_old_mine` ranks a randomly chosen mine cell nearer to a past event than a randomly chosen clear cell 91.9% of the time, against 67.2% for the next-best feature, and 384 of the 430 mine cells (89.3%) lie within 1 km of a recorded event, compared with 20.9% of clear cells. This fits how mines are laid. In my experience, mines are generally placed in groups to achieve a tactical or strategic aim, and those placed at random are generally there for nuisance or

to protect something. Military operations employ minefields for several reasons: to block, fix, turn or disrupt an enemy's movement, the four effects of tactical obstacles (Xenos, n.d.), and to protect positions. Whether mines protect an area or serve a military aim, they are laid as a group, so a recorded event points to others nearby. In Colombia, illegal armed groups laid mines "to protect areas where they grow illegal crops, to hamper the advance of security forces or to protect areas that they consider to be of strategic interest" (Republic of Colombia, 2010, p. 1), and the RELand study rests on the same point, that mines are used according to the logic of the war rather than at random (Dulce Rubio et al., 2024).

Table 1

Scores of the Six Rankings on the 15 Held-Out Municipalities

Ranking	PR-AUC, mean (SD)	PR-AUC, 12 municipalities	ROC-AUC, mean (SD)	Top 10%: recall	Top 10%: precision
Random ranking	.021 (.023)	.026	.453 (.167)	.067	.011
Distance rule	.194 (.115)	.215	.864 (.115)	.616	.099
Logistic regression, 63 features	.213 (.233)	.183	.854 (.155)	.591	.095
Gradient-boosted trees, 63 features	.218 (.228)	.189	.782 (.213)	.535	.086
Logistic regression, 62 features	.053 (.063)	.063	.629 (.200)	.233	.037
Gradient-boosted trees, 62 features	.109 (.246)	.052	.690 (.152)	.170	.027

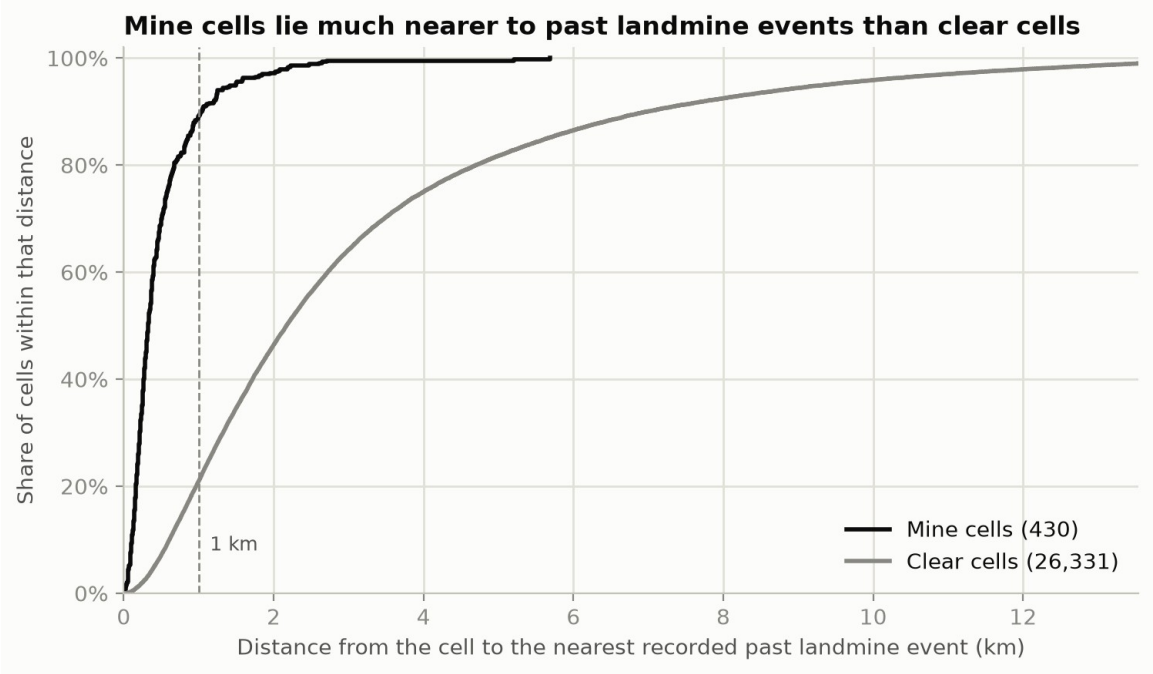
Note. From notebook Section 7.1. PR-AUC is average precision. Means and standard deviations are over the 15 held-out municipalities; the "12 municipalities" column leaves out the three with only one or two mine cells. Top 10% pools each municipality's 10% highest-ranked cells (2,669 of 26,761 cells; 430 mine cells in all). The 62-feature models leave out `dist_old_mine`.

Read as a survey plan (Figure 2), the rankings with `dist_old_mine` found most mine cells early. Surveying the top 10% of each municipality's cells found 61.6% of all mine cells with the distance rule, 59.1% with the logistic regression and 53.5% with the trees. Deeper in the ranking the logistic regression overtook the rule: after 20% of the cells it had found 80.5% of the mine cells, against 78.8%, and after 30% it had found 92.1%, against 87.7%. The trees stayed

behind both. Without `dist_old_mine`, both models fell far behind the rule, finding 23.3% and 17.0% of the mine cells in the top 10%; that is more than the random order's 6.7%, so the other 62 features carry some signal, but most of what the full models knew came from the distance to past events.

Figure 1

Distance From Mine Cells and Clear Cells to the Nearest Recorded Past Landmine Event



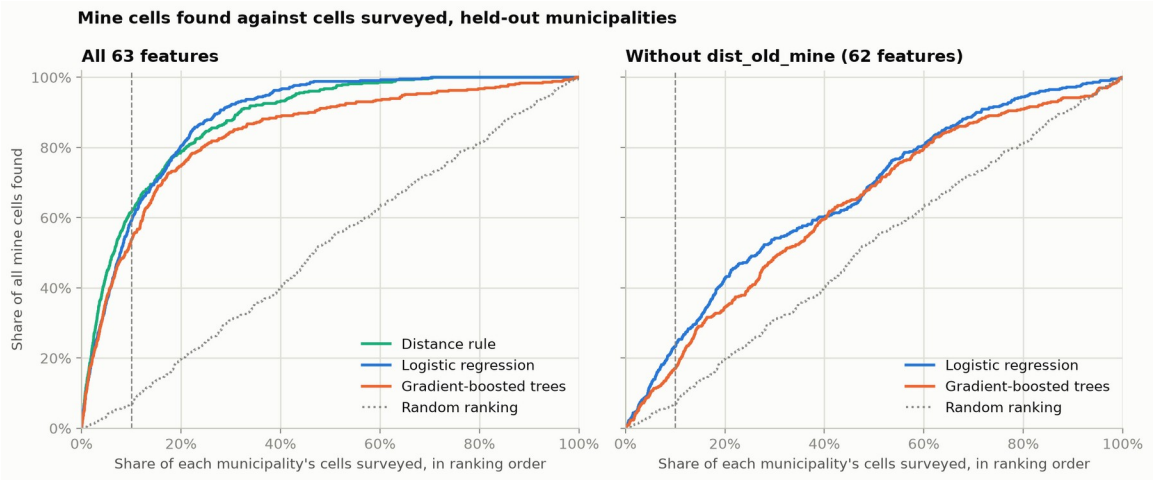
Note. From notebook Section 3.2. Each line shows the share of cells that lie within a given distance of the nearest recorded past landmine event; the dashed line marks 1 km. Data from mdulcer10 (2026).

The scores varied more between municipalities than between rankings (Figure 3). Each model scored higher than the distance rule in 6 of the 15 municipalities and lower in the other 9; in San Francisco, for example, the rule scored .384, against .057 for the logistic regression and .166 for the trees. The three municipalities with one or two mine cells show how fragile a score on so few cells is: in La Unión one cell decided the result, while in Chigorodó and San

Roque every ranking put the mine cells hundreds of places down, so every score there was close to zero.

Figure 2

Mine Cells Found Against Cells Surveyed, on Held-Out Municipalities



Note. From notebook Section 7.4. Each curve surveys every municipality's cells in the order of one ranking; the dashed line marks the top 10% of cells. Left: rankings that use `dist_old_mine`; right: the models trained without it.

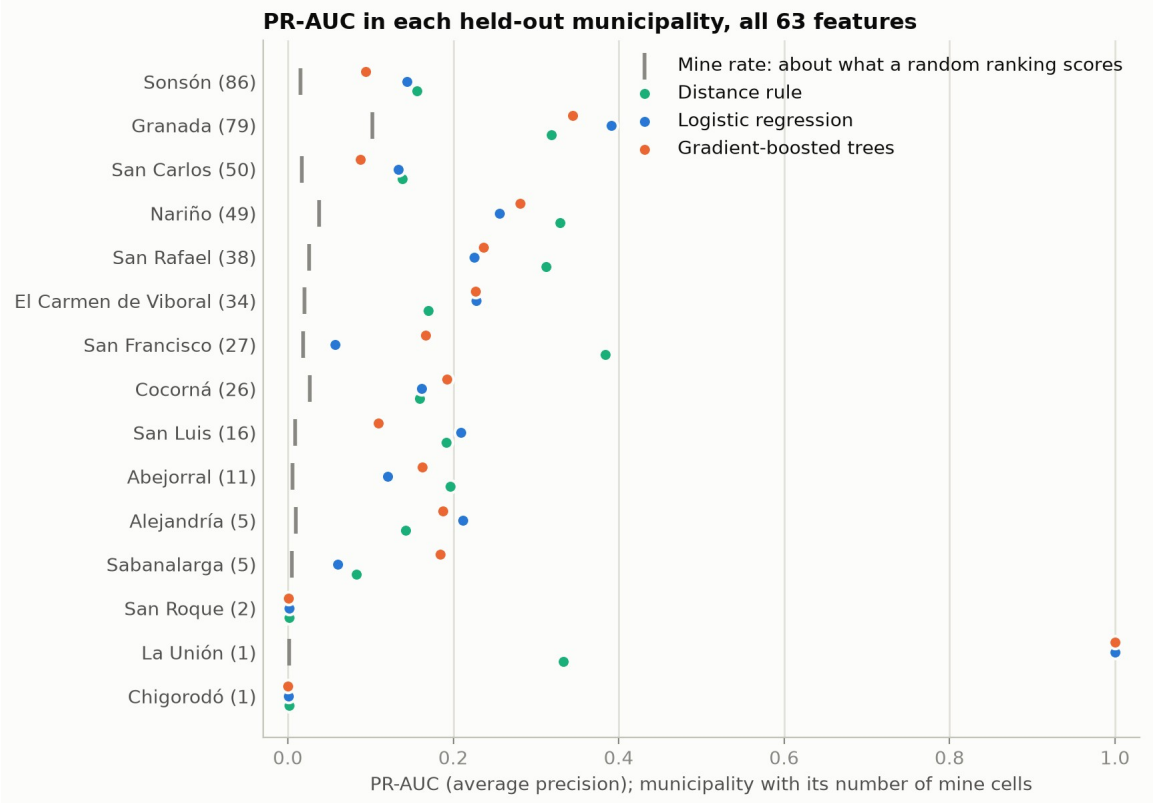
The logistic regression's weights show the same dependence on past events (Figure 4). Fitted on all 15 municipalities, it put a weight of −1.46 on the distance to past events, more than four times its next largest weight (0.35, the distance to an airport), so nearer a past event meant a higher score; most of the other weights were small and described remoteness, with higher scores for cells farther from airports, seaports, roads and buildings and for cells with fewer people.

Two checks put these results in context. First, a random split would have told a different story: on random test cells, the trees' mean PR-AUC within municipalities was .386, about twice the .190 they scored on the same municipalities when each was held out whole, and the logistic regression's was .258 against .194, while the untrained distance rule moved only from .227 to .246 (notebook Section 8). Second, the distance rule reproduced the single-feature logistic

regression of the RELand study to two decimals, with a ROC-AUC of 86.35 and a PR-AUC of 17.07 on the paper's scale of 100, which supports both the data and the way I scored them (Dulce Rubio et al., 2024). The trained models scored below their counterparts in the paper on ROC-AUC (85.40 against 86.54 for the logistic regression, and 78.25 against 88.85 for gradient boosting); one likely reason is that the paper's models had six more features describing past events (notebook Section 7.2).

Figure 3

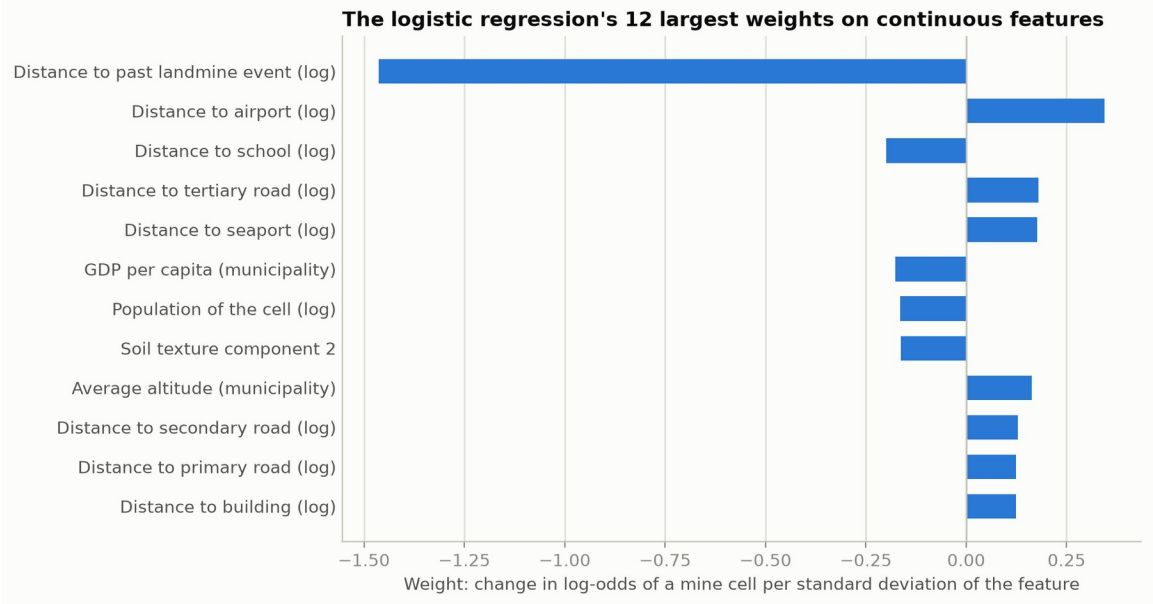
PR-AUC in Each Held-Out Municipality



Note. From notebook Section 7.5. All 63 features. Each row is one held-out municipality, with its number of mine cells in parentheses; the grey tick marks the municipality's mine rate, about what a random ranking scores there.

Figure 4

The Logistic Regression's 12 Largest Weights on Continuous Features



Note. From notebook Section 9. Fitted on all 15 municipalities with `C` = 0.001. A weight is the change in the log-odds of a mine cell for one standard deviation of the feature, after the log where one is applied; the weights describe associations in the data, not causes.

Interpretation for a Non-Technical Audience

In plain terms, the computer models did not beat a simple rule of thumb. If a team surveyed first the tenth of each municipality's land that lies nearest to places where landmine accidents or events had been recorded, it would have reached about six in ten of the places where mines were later found; the models did about as well, but no better. In that first tenth, about one cell in ten held mines, against fewer than two in a hundred across all the land, so the ranking would have made the first visits about six times as productive as a random choice. Without the record of past events, the models found only about a fifth of the mined places in the same tenth of the land, which shows how much the history of accidents tells us and how little the other open data add. The models also looked far better when tested the easy way, on cells mixed at random,

than on whole municipalities they had never seen, so a model judged the easy way would be trusted more than it deserves.

For me, the value of a tool like this is as a digital survey: an information tool that helps to plan even the surveys more efficiently, before any team is sent. Survey is the quicker task, much like a reconnaissance, and in land release survey normally takes priority over clearance (United Nations Mine Action Service [UNMAS], 2026). Survey teams can gather information relatively quickly, before the infrastructure is moved in to support the main body for deliberate operations, and this ensures that the proper resources go to the right areas. Clearance can use heavy equipment and specialised equipment such as metal detectors and blast equipment, and it needs a presence on site for long periods, relationships with the community, local hospitals informed that deliberate operations are under way, and detailed planning. A ranking would give us target hotspots to start from, but we would still need to thoroughly vet the areas we send teams into. If we can clear land more efficiently and release it to people sooner, that enables the economic growth that comes afterwards. In Colombia, comprehensive humanitarian demining that involved affected communities increased local economic activity and population density, especially where market access was better (Prem et al., 2025), while in Mozambique, economic activity did not pick up when operators investigated areas that earlier surveys had mistakenly marked as contaminated (Chiovelli et al., 2025): effort that a better ranking could save.

Putting those resources in the right places first matters most where contamination is vast. In April 2023, GLOBSEC estimated that if all 174,000 square kilometres of Ukrainian territory that had seen combat were contaminated, the demining capacity then available would need more than 700 years to clear it: 757 years in the report's summary and 737 years in its needs assessment (Osmolovska, 2023). In my view that estimate should be projected forward, because

mines have continued to be laid since then: in 2026, Humanity & Inclusion reported that anti-personnel mines "have also been laid in significant numbers, and this trend is increasing" (Humanity & Inclusion, 2026). Survey has narrowed the area at risk at the same time, to about 132,076 square kilometres in February 2026, with more than 41,924 square kilometres returned to safe use (Humanity & Inclusion, 2026), and that is the kind of work a digital survey could help to plan. If this problem set can be refined, I believe it may help to accelerate the pace of clearance there.

Limitations and Potential Bias

Limitations of the Data and Models

- **One region, few mine cells.** All 15 municipalities are in Antioquia, and three hold only one or two mine cells, so their scores rest on one or two cells; La Unión alone moved the models' mean PR-AUC above the rule's. The results may not hold in other regions or in other kinds of conflict. In my experience, context is everything: professional militaries typically lay minefields in strategic and tactical patterns, which are likely easier to map, while non-state armed groups may not be trained in such a way and place mines where you would not expect them. The mix of mines differs too: in military operations, anti-personnel mines often protect anti-tank mines, because bigger obstacles are needed to stop bigger vehicles, while anti-personnel mines on their own typically cut off footpaths or areas where infantry would move, and can also protect drug-growing areas. Colombia's mines were laid by such groups, which chose where to lay them for a purpose but placed them "randomly and indiscriminately, without regard for any military doctrine" (Republic of Colombia, 2010, p. 3). A ranking learned in Antioquia may

therefore not carry over to a conventional war such as Ukraine's, and I found no published study that has tested such a transfer.

- **Labels show where mines were found.** A cell is positive if demining found mines there, so the labels reflect where demining took place as well as where mines lay. A clear cell is not proof that no mines remain either. In my experience, in a former conflict area everything in the end needs to be cleared: building sites in Germany and other places still need to be cleared of ordnance before digging, eight decades after the Second World War. Land release accepts a residual risk, the "risk that remains despite having applied all reasonable effort during the land release process" (UNMAS, 2026, Section 3.18). If some cells labelled clear still hold mines, a ranking that puts them near the top is counted as wrong there, so its scores would be too low.
- **A feature that may share the labels' source.** `dist_old_mine` comes from the same national mine action records as the labels, and I found no statement on whether the recorded events came before the demining behind the labels (Dulce Rubio et al., 2024). If some events were recorded during that demining, part of the strength of the distance rule, and of the full models, restates the labels rather than predicting them.
- **Fewer features than the study.** This release has 63 of the paper's 70 features, and six of the seven missing ones describe past events, so the comparison with RELand is approximate.
- **Unstable tuning.** With only 14 municipalities to tune on, the chosen settings changed from round to round, which adds noise to each round's result.
- **A trade-off in preprocessing.** Using one log transformation for every distance kept the model simple but gave seven barely skewed features a left skew.

Potential Bias

- **Reporting depends on who reports.** In my experience the bias generally comes with the skills of the people within the region. Once United Nations crews or professional deminers are deployed, the issue is generally alleviated, but before trained professionals arrive, the reports can be all over the place. This applies most to the records of past events behind `dist_old_mine`, which the models treat as reliable; the labels come from completed humanitarian demining.
- **Learning the records rather than the mines.** Because the strongest feature is the record of past events, a model can learn where events were recorded, which may differ from where mines lie, and it will rank poorly the places where nothing was ever recorded.
- **Who waits longest.** Survey order also follows land use. In my experience, the immediate priority is always farmland and land that will be built on soon, while the areas generally surveyed last are those people already commonly use, those that had some distance to the conflict or can be shown not to have been involved in it, and jungle and rocky terrain of little economic value, so people who depend on such land may wait longest. That land may be cleared only when construction requires it. In Canada, for example, ordnance is fired in some mountain passes to trigger controlled avalanches (Department of National Defence, 2026), and in my experience any construction there needs the area cleared and an explosive ordnance disposal (EOD) technician on site. A risk ranking is one input to that order, not a replacement for it.

Misuse and the Step to Reduce It

The main risk lies in how a ranking is used. In my experience, it could lead to an inefficient application of demining resources, and if an area is under-prioritised and, for some

reason, released as safe, this could cause injuries and accidents. A low score is not evidence of safety: land release removes all presence or suspicion of explosive ordnance through non-technical survey, technical survey and/or clearance (UNMAS, 2026), not through a prediction. The step I propose, in line with the dataset's own warning that predictions should not be "the sole basis for demining operations without expert human review and validation in the field" (mdulcer10, 2026), is that the model is never used on its own but only as an efficiency aid that orders survey, and that no land is cancelled or released on a model score without survey evidence. Before a model set the order of survey in the field, it would have to show a reasonable confidence level and then be tried for a period, demonstrating that deployment teams were actually more efficient. This matches the view that any safety-related use of machine learning in mine action "should be subjected to rigorous trials" judged by independent specialists in both mine action and AI (Gasser, 2024, Conclusions section). The steps I took in the analysis reduce a related risk, overconfidence: I tested the models on municipalities they had never seen, reported the results for each municipality, trained them again without `dist_old_mine`, and showed how much a random split would have flattered them.

Next Steps

With more time and data, I would overlay drone data, satellite feeds and other forms of sensor and cross-correlate them with the model's predictions to verify their accuracy on the ground, and I would do this for all regions of the world. Training a model in one country and testing it in another would show whether it transfers beyond the conflict it learned from. Records of when each past event was reported, compared with when survey took place, would settle whether `dist_old_mine` predicts the labels or restates them.

Citation Requirements

I used citations for machine learning concepts and tools, for the choice of models, testing protocol and metrics, and for the limitations and risks of use, but not for my own results, which come from the notebook, or for points from my own experience in mine action, which I state as such. Dulce Rubio et al. (2024) and the dataset card (mdulcer10, 2026) support the data, the testing protocol and the benchmark; Roberts et al. (2017) support testing on held-out municipalities; Saito and Rehmsmeier (2015) support the choice of PR-AUC; Pedregosa et al. (2011), Ke et al. (2017) and the scikit-learn documentation support the models; UNMAS (2026) supports the points on land release and residual risk; the Republic of Colombia (2010) supports who laid Colombia's mines and why; Prem et al. (2025) and Chiovelli et al. (2025) support the economic effects of demining; Gasser (2024) supports trials before field use; the Department of National Defence (2026) supports the example of avalanche control; Xenos (n.d.) supports the military aims of minefields; and Osmolovska (2023) and Humanity & Inclusion (2026) support the figures for Ukraine. The references follow APA 7.

I used Claude (Anthropic). I ran the notebook and checked its results.

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